AI CLASS · PROJECT PROPOSAL

AI Model *Lab* · AI tools used

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Live project: https://model-lab-library.vercel.app
Presentation: https://model-lab-library.vercel.app/present
Source code: https://github.com/phonesett10401/model-lab

AI MODELS IN THE LAB

YOLO26 Nano (Ultralytics)	Sea creature detector (object detection)	github.com/ultralytics/ultralytics
EfficientAT (MobileNet mn10)	Sound detective (audio tagging)	github.com/fschmid56/EfficientAT
Qwen2.5-1.5B-Instruct	Writes the RAG assistants' answers	huggingface.co/Qwen/Qwen2.5-1.5B-Instruct
snowflake-arctic-embed-s	Finds the right document passages for the RAG assistants	huggingface.co/Snowflake/snowflake-arctic-embed-s

AI FRAMEWORKS FOR TRAINING AND RUNNING

PyTorch	Training both models	pytorch.org
Ultralytics	Training and evaluating the detector	docs.ultralytics.com
ONNX / ONNX Runtime Web	Running the two models in the browser, on the device	onnxruntime.ai
WebLLM (MLC)	Running the RAG assistants in the browser	github.com/mlc-ai/web-llm

AI USED WHILE PREPARING THE DATA

OpenCLIP	Photo-data pipeline (filtering step)	github.com/mlfoundations/open_clip
AudioSet-pretrained model (EfficientAT)	Double-checked extra sound clips before training	research.google.com/audioset

AI ASSISTANT USED TO BUILD THE PROJECT

Claude Code (Anthropic)	Helped write the site, training scripts and presentation deck	claude.com/claude-code

DATASETS THE MODELS LEARNED FROM

Open Images V7 (CC BY 2.0)	Sea creature photos	storage.googleapis.com/openimages/web
Aquarium Combined (Roboflow, CC BY 4.0)	Sea creature photos	universe.roboflow.com
FSD50K and Freesound (CC0 / CC BY)	Sound clips	zenodo.org/records/4060432 · freesound.org